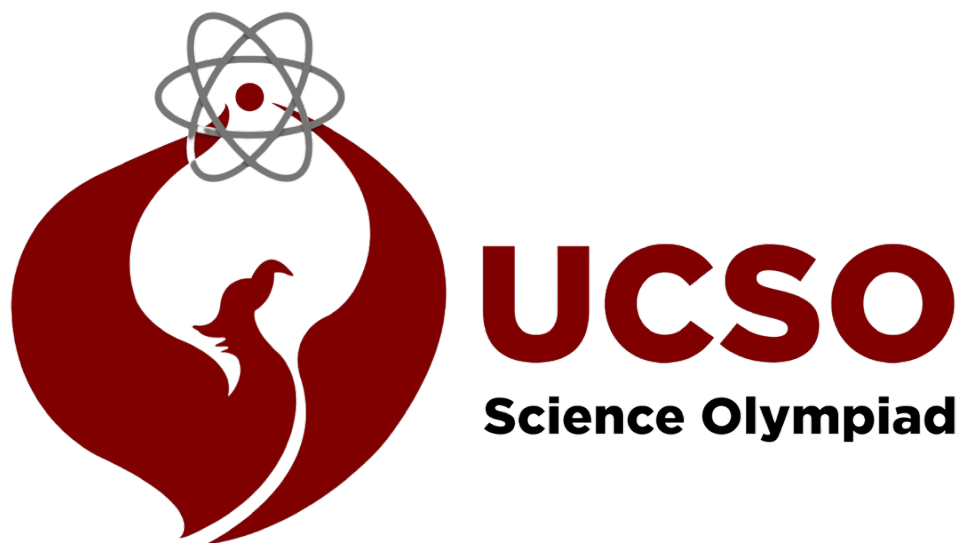


Dynamic Planet — Oceanography
Test Document



2026 University of Chicago Science Olympiad Invitational — Div C

****DO NOT WRITE ON TEST DOCUMENT****

By Josh Gonzalez and Matthew Castillo (UC CO'27)

1 Multiple Choice (40 Points)

Select the appropriate answer option (4pts/Question)

1. Which of the following best describes cabbeling?
 - A. The formation of sea ice resulting in brine rejection and increased salinity
 - B. The mixing of water masses of identical density but different temperatures and salinities, resulting in a mixture that is denser than either parent mass
 - C. The non-linear relationship between pressure and depth due to the compressibility of seawater
 - D. The conservation of potential vorticity leading to western intensification of gyres
2. In the context of Geostrophic flow, if the isopycnals are tilted downward toward the East in the Northern Hemisphere, what is the direction of the resulting geostrophic current?
 - A. North
 - B. South
 - C. East
 - D. West
3. Which term represents the ratio of inertial forces to coriolis forces, used to determine if geostrophic balance is an appropriate approximation for a flow?
 - A. Reynolds number (Re)
 - B. Froude number (Fr)
 - C. Rossby number (Ro)
 - D. Ekman number (Ek)
4. Considering the biological pump, which process is primarily responsible for the remineralization of organic matter in the mesopelagic zone?
 - A. Photosynthetic fixation
 - B. Heterotrophic respiration and bacterial decomposition
 - C. Calcium carbonate dissolution below the CCD
 - D. Anthropogenic CO_2 uptake
5. Which tidal constituent typically has the largest amplitude in most ocean basins and represents the principal lunar semidiurnal tide?
 - A. K_1
 - B. M_2
 - C. S_2
 - D. O_1
6. The Sverdrup Balance relates meridional transport in the ocean interior to:
 - A. The curl of the wind stress
 - B. The gradient of sea surface temperature
 - C. The bottom friction coefficient
 - D. The horizontal density gradient
7. In an Amphidromic system, cotidal lines radiate outward from the amphidromic point. What does the value associated with a cotidal line represent?
 - A. The tidal range at that location
 - B. The speed of the tidal current
 - C. The phase of the tide (time of high water)
 - D. The resonance frequency of the basin

8. Why is the Brunt-Väisälä frequency (N) roughly zero in the mixed layer?
- A. Because the density gradient is vertical
 - B. Because the fluid is vertically unstratified (homogeneous density)
 - C. Because salinity compensates for temperature changes exactly
 - D. Because the Coriolis force is negligible at the surface
9. Which sediment type would you most likely find accumulating below the calcium carbonate compensation depth (CCD)?
- A. Pteropod ooze
 - B. Foraminiferal ooze
 - C. Siliceous ooze
 - D. Evaporites
10. During an El Niño event, the flattening of the thermocline across the equatorial Pacific is associated with:
- A. A strengthening of the Walker Circulation
 - B. A breakdown of the geostrophic balance
 - C. The propagation of an equatorial Kelvin wave transporting warm water eastward
 - D. Intensified upwelling off the coast of Peru

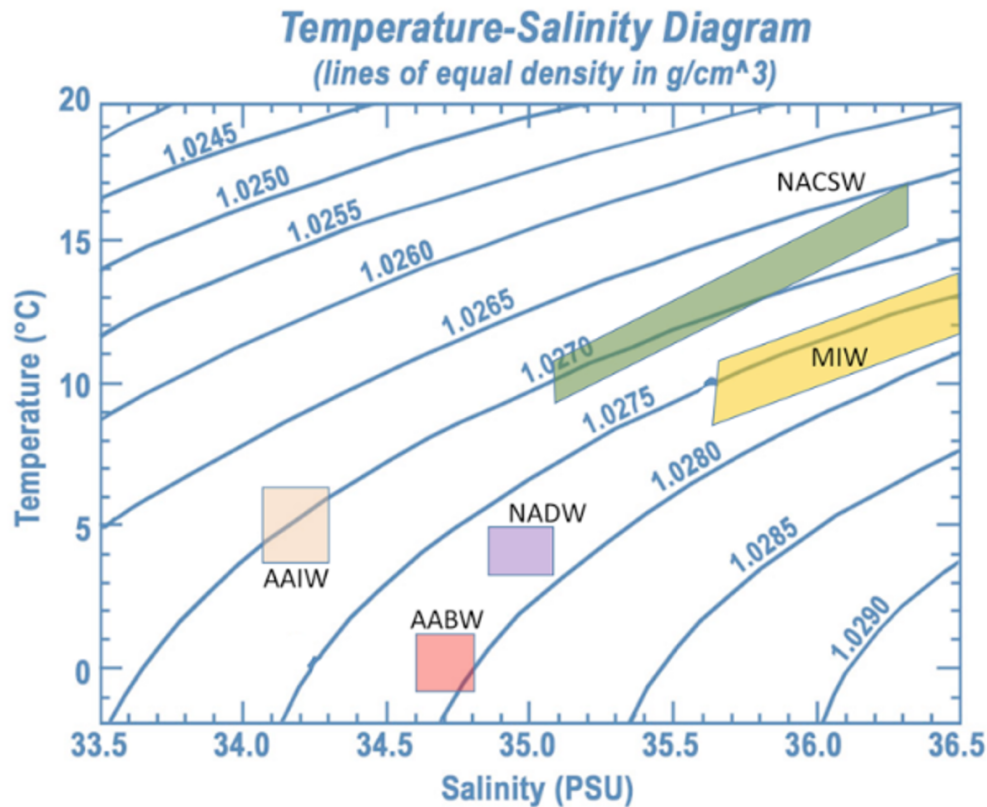
2 Short Answer (50 Points)

Answer the following in 2-3 sentences. Equations may be used as part of answers (5pts/Question)

1. Explain how the curl of the wind stress ($\Delta \times \tau$) leads to Ekman pumping or suction. specifically, what vertical velocity (w_E) results from a cyclonic wind stress pattern in the Northern Hemisphere?
2. Why do oceanographers prefer Potential Temperature (θ) over in-situ Temperature (T) when analyzing deep-water stability? Briefly describe the thermodynamic reason T changes when a water parcel is moved adiabatically to a different depth.
3. Define the Beta Effect (β) in physical terms. How does the variation of the Coriolis parameter (f) with latitude contribute to the conservation of potential vorticity?
4. Describe the "Knudsen relations" (salt and volume conservation) for a two-layered estuary. If river discharge into the estuary decreases, how must the inflowing seawater volume change to maintain constant salinity?
5. How does the ratio of Oxygen-18 to Oxygen-16 ($\delta^{18}\text{O}$) in benthic foraminifera shells serve as a proxy for global ice volume and deep-ocean temperature?
6. Explain the difference between Stokes' Law settling and turbidity currents. Why does Stokes' Law fail to predict the transport of sediment to the abyssal plains?
7. Stommel's model attributes western intensification to the variation of the Coriolis effect with latitude. Explain physically why the western boundary current must be narrow and fast compared to the eastern return flow to conserve vorticity.
8. What condition must be met for a semi-enclosed basin, such as the Bay of Fundy, to experience tidal resonance? Reference the natural period of oscillation in your answer.
9. Differentiate between "salt fingers" and "diffusive convection." Which regime occurs when warm, salty water overlies cold, fresh water?
10. Why is the North Pacific incapable of forming deep water (like NADW), despite having similar high latitudes to the North Atlantic?

3 Diagram Analysis (50 Points)

Answer the following in 2-4 sentences. Equations may be used as part of answers (10pts/Question)



1. Identify the region on the T-S diagram representing the pycnocline. Explain how you can determine the static stability of the water column ($E = -\frac{1}{\rho} \frac{d\rho}{dz}$) purely by looking at the vertical spacing of data points relative to the isopycnals.
2. Trace the mixing line between AAIW and NADW. If a water parcel consists of 60% NADW and 40% AAIW, calculate its approximate resulting salinity using the Lever Rule approximation (Assume $S_{\text{NADW}} = 34.9$, $S_{\text{AAIW}} = 34.4$).
3. Explain the significance of the curvature of the plot's isopycnals (σ_t) in relation to the Thermal Expansion Coefficient (α) and how it varies with temperature.
4. Which water parcel has the least uniform density?
5. If a T-S profile shows a sharp cusp/kink rather than a smooth curve, what physical oceanographic feature is likely being crossed? Explain this in terms of interleaving water masses.

4 Long Response Questions (30 Points)

Answer the following in detail. Equations/calculations must be used as part of answers (15pts/Question)

1. Derive the equations for geostrophic balance starting from the horizontal momentum equations (Navier-Stokes) on a rotating frame. Explain why geostrophic currents flow parallel to isobars rather than down the pressure gradient. You may:
 - Assume steady state ($\frac{\partial u}{\partial t}$) and negligible friction $F_x = F_y = 0$
 - Show that the zonal (u) and meridional (v) velocities are proportional to the slope of the sea surface height (η)
2. Compare and Contrast Rossby Waves and Kelvin Waves in the ocean.
 - (i) Identify the primary restoring force for each wave type.
 - (ii) Contrast their propagation directions.
 - (iii) Explain the role of the coastal boundary in the existence of a Kelvin wave and why its amplitude decays exponentially offshore.

5 Physical Reasoning Question (30 Points)

The SOund Fixing And Ranging (SOFAR) Channel is a deep ocean layer where low-frequency sound travels vast distances efficiently. The speed of sound in seawater is given by the Newton-Laplace equation: $c = \sqrt{\frac{K}{\rho}}$, where K is the bulk modulus and ρ is density.

- (i) How might the bulk modulus (K) and Density (ρ) depend on temperature and depth?
- (ii) Mathematically demonstrate why a minimum sound speed exists at approximately 1000m depth. Specifically, discuss the interplay between the thermal effect on compressibility ($\frac{\partial c}{\partial T}$) and the pressure effect on bulk modulus ($\frac{\partial c}{\partial P}$) in the upper vs. deep ocean. Use the coordinate z for depth.
- (iii) Explain how this minimum allows for long-distance propagation of longitudinal sound waves without significant attenuation.